# Dan Schumacher

PhD Student, Information Technology

*One UTSA Circle, San Antonio, TX 78249*

daniel.schumacher@utsa.edu | https://github.com/danschumac1

## EDUCATION

**UT San Antonio** — *San Antonio*
PhD in Information Technology — *Aug 2024 - present*
- Advisor: Dr. Anthony Rios

**UT San Antonio** — *San Antonio*
MS in Data Analytics — *Aug 2023 - July 2024*

**UT San Antonio** — *San Antonio*
BM in Guitar Performance — *Aug 2014 - May 2018*

**Domain:** Natural Language Processing; Human-AI Collaboration; Mechanistic Interpretability; AI Applications; Generative AI in Games; AI in Healthcare.

## PROFESSIONAL EXPERIENCE

**UT San Antonio** — *San Antonio, TX*
Graduate Research Assistant — *2023 - Present*
- Conducted AI applications, AI reasoning, AI security, and AI education research.

**Pieces Technologies (now SmarterDx)** — *San Antonio, TX*
AI Engineer Intern — *2024*
- Designed and deployed a data processing pipeline that transformed structured healthcare data into clinically meaningful events and indicators.
- Collaborated with clinical experts to translate complex medical definitions and triggering conditions into robust, automated workflows.

**LJE Music, Self Employed, and UTSA String Project** — *Various Locations*
Music Educator — *2016 - 2024*
- Communicated with hundreds of student-parents.
- Wrote orchestral arrangements of popular music from video games and movies.
- Led teacher ensembles in elementary school concerts to increase enrollment.
- Conducted orchestra of 180 young musicians.
- Taught group music classes for K-8th graders.
- Tracked instrument rentals and attendance.

**Self Employed** — *Various Locations*
Classical Guitarist — *2016 - 2024*
- Biweekly performances at the San Antonio International Airport.
- Performed for weddings, grand openings, and a nun convention.
- Performed in concert tour throughout Cuba, and donated guitar strings to local schools.

**San Antonio Public Library** — *San Antonio, TX*
Librarian Assistant — *2021 - 2023*
- Developed resources for phone-less patrons using Text Now.
- Assisted a patron in making their first phone call to their mother in over a year.
- Restocked Central Library's banned book collection.

## PUBLICATIONS

1. **Detecting AI Imposters: How Do Middle Schoolers Identify LLM Agents in a Live Collaborative Setting?**
   **Dan Schumacher**,Pragathi Durga Rajarajan, Haven Kotara, Roman Rendon, Kosi Atupulaz, Deepti Tagare, Ismaila Sanusi, Fred Martin, and Anthony Rios
   In: Main conference of the Empirical Methods in Natural Language Processing 2026: **EMNLP 2026**.

2. **Prompting Underestimates LLM Capability for Time Series Classification.**
   **Dan Schumacher**, Erfan Nourbakhsh, Rocky Slavin, and Anthony Rios
   In: Main conference of the 15th International Joint Conference on Natural Language Processing **IJCNLP-AACL 2026**.

3. **Telling Speculative Stories to Help Humans Imagine the Harms of Healthcare AI.**
   Xingmeng Zhao, Tongnian Wang, **Dan Schumacher**, Veronica Rammouz, and Anthony Rios
   In: Findings of the Association for Computational Linguistics: **ACL 2026**.

4. **AI for Everyone: Engaging Middle Schoolers through Collaborative, Ethical, and Multimodal AI Learning.**
   Kayleigh Stallings, Nicole Tian, Elif Yayla Ercek, Haven Kotara, Devin Marinelli, Pragathi Durga Rajarajan, **Dan Schumacher**, Ismaila Temitayo Sanusi, and Fred Martin
   In: Proceedings of the 57th ACM Technical Symposium on Computer Science Education (**SIGCSE TS 2026**), New York, NY, USA, 2026.

5. **RASTeR: Robust, Agentic, and Structured Temporal Reasoning.**
   **Dan Schumacher**, Fatemeh Haji, Tara Grey, Niharika Bandlamudi, Nupoor Karnik, Gagana Uday Kumar, Cho-Yu Jason Chiang, Peyman Najafirad, Nishant Vishwamitra, and Anthony Rios
   In: Proceedings of the 14th International Joint Conference on Natural Language Processing (**IJCNLP-AACL 2025**), Mumbai, India, 2025.

6. **Charting the Future: Using Chart Question-Answering for Scalable Evaluation of LLM-Driven Data Visualizations.**
   James Ford, Xingmeng Zhao, **Dan Schumacher**, and Anthony Rios
   In: Proceedings of the 31st International Conference on Computational Linguistics (**COLING 2025**), Abu Dhabi, UAE, 2025.

7. **Bike Frames: Understanding the Implicit Portrayal of Cyclists in the News.**
   Xingmeng Zhao, **Dan Schumacher**, Sashank Nalluri, Suhana Shrestha, Xavier Walton, and Anthony Rios
   In: Proceedings of the International AAAI Conference on Web and Social Media (**ICWSM 2025**), June 2025.

8. **Team UTSA-NLP at SemEval 2024 Task 5: Prompt Ensembling for Argument Reasoning in Civil Procedures with GPT-4.**
   **Dan Schumacher** and Anthony Rios
   In: Proceedings of the 18th International Workshop on Semantic Evaluation (**SemEval-2024**), Mexico City, Mexico, 2024.

### Publications Under Review

9. **Dementia vs. General Caregiver: A Large-Scale Comparison of Support Needs using Caregiving Communities on Reddit.**
   **Dan Schumacher**, Xingmeng Zhao, Visha Arumugam, Sashank Nalluri, John Quarles, Jeremy Davis, and Anthony Rios

T1. **Webside Manner: A Dataset and Analysis of Patient-Centered Communication Strategies in Verified Clinician Responses.**
   Harley Yanlai Fan, Xingmeng Zhao, **Dan Schumacher**, Yifan Gao, and Anthony Rios
   ACL ARR 2026 August Submission, Aug. 2026

T2. **Understanding Clinical Cognitive Dialogues Using Large Language Models.**
   Vishalakshi Arumugam, **Dan Schumacher**, Veronica Rammouz, Erfan Nourbakhsh, Enrique Gonzalez, Jeremy J Davis, and Anthony Rios
   AAAI 2027 Artificial Intelligence for Social Impact Track Submission, Jul. 2026

## Awards, Fellowships, & Grants

2017 **String Project Excellency Award**, UTSA College of Liberal and Fine Arts
2017 **Outstanding Achievement in Guitar Performance**, UTSA College of Liberal and Fine Arts
2018 **Outstanding Achievement in Guitar Performance**, UTSA College of Liberal and Fine Arts

## TALKS & PRESENTATIONS

T3. **Prompting Underestimates LLM Capability for Time Series Classification.**
   Oral Presentation at Cybersecurity and Artificial Intelligence Fall 2026 Review Meeting, UT San Antonio, Sep. 2026

T4. **Detecting AI Impostors: How Do Middle Schoolers Identify LLM Agents in a Live Collaborative Setting?**
Poster Presentation at Cybersecurity and Artificial Intelligence Fall 2026 Review Meeting, UT San Antonio, Sep. 2026

T5. **Prompting Underestimates LLM Capability for Time Series Classification.**
Oral Presentation at Cybersecurity and Artificial Intelligence Fall 2026 Review Meeting, Texas A&M University , Sep. 2025

T6. **Garbage In, Not (Always) Garbage Out: Robust Temporal Reasoning with RASTeR.**
Brown Bag Seminar Series, Department of Information Systems and Cybersecurity, UT San Antonio, Nov. 2025.

T7. **RASTeR: Robust, Agentic, and Structured Temporal Reasoning.**
Oral Presentation at the 14th International Joint Conference on Natural Language Processing (**IJCNLP-AACL 2025**), Mumbai, India, Dec. 2025.

T8. **RASTeR: Robust, Agentic, and Structured Temporal Reasoning.**
2025 Fall Workshop on Cybersecurity and Artificial Intelligence, Angelo State University, San Angelo, TX, Oct. 2025.

T9. **Prompting Underestimates LLM Capability for Time Series Classification.**
Poster Presentation at the **Texas NLP Symposium**, A&M College Station, 2025.

T10. **Context Matters: An Empirical Study of the Impact of Contextual Information in Temporal Question Answering Systems.**
Poster Presented at the Los Datos Conference 2024, San Antonio, TX, 2024.

T11. **Presenter for AITX.**
Boeing Center at Tech Port, San Antonio, TX.

## TEACHING EXPERIENCE

2026-2027 **UT San Antonio**, Instructor
- IS2053: Introduction to Programming.
- Class Size: 40
- Foundational Python for cyber security students.

2016-2024 **LJE Music and Self Employed**, Private Lesson Instructor
- Class Size: 1
- Singer songwriter, classical and electric guitar, piano
- Ages 4 to 75

2022-2023 **Digital Help Hour**, Librarian Assistant
- Class Size: Varies
- One on one sessions for the public in need of technology assistance
- Largely worked with people who are homeless get set up with computers/tablets/phones

2015-2018 **UTSA String Project**, Assitant Director, Guitar Coordinator
- Class Size: 20
- Group classical guitar lessons
- grades K-8

## MENTORSHIP EXPERIENCE

Gabriel Green, (Spring 2024), UTSA MSIT Student

## PROFESSIONAL ACTIVITIES

**Reviewer** ACL Rolling Review (2024–2026) | Dewald Roode Workshop on Information Systems Security Research

## RELEVANT EXPERIENCE AND SKILLS

S1. Deep Learning: Enhanced Model's temporal reasoning ability. Made more robust to irrelevant context.

S2. Large Language Models and Agentic Systems Research Seminar: Investigated demographic biases in large language models in game theory settings.

S3. Natural Language Processing Research Seminar: Discovered security vulnerabilities caused by the growing context windows in large language models.

S4. Game Development: Made a dog agility platform puzzle game with a LLM-powered haiku-writing level up system.